

# Case Analysis – Avoiding Misleading Proportions

## Question 1: Truncated Y-Axis in Sales Growth

### 1. Why is the chart misleading?

The Y-axis starts at **90 instead of 0**, so the visible bar heights represent only the range **90–105**.

- January: 95 → visible height = 5
- June: 105 → visible height = 15

The chart makes June appear **3 times taller** than January, even though the actual increase is only:

- $105 - 95 = 10$  lakhs
- Percentage increase =  $10/95 \times 100 \approx 10.5\%$

Thus, the chart exaggerates a small increase.

### 2. Violated visualization principle

**Proportional representation / truthful scale principle** (bars should start from zero so lengths represent values accurately).

### 3. Improved visualization

- Use a **bar chart with Y-axis starting at 0**, or
- Use a **line chart** to show monthly sales trends without exaggerating differences.

### 4. Possible influence on business decisions

Managers may incorrectly believe sales growth is dramatic and may:

- Increase marketing budgets unnecessarily,
- Expand inventory too aggressively,
- Overestimate demand and revenue forecasts.

## Question 2: Pie Chart with Incorrect Proportions

Total students =  $420 + 280 + 200 + 100 = 1000$

### 1. Correct percentages

| Department       | Students | Percentage |
|------------------|----------|------------|
| Computer Science | 420      | 42%        |
| Electronics      | 280      | 28%        |
| Mechanical       | 200      | 20%        |
| Civil            | 100      | 10%        |

## 2. Why is the chart misleading?

The chart shows approximately **50%, 30%, 15%, 15%**, which does not match the data.

- Computer Science is overstated (50% vs 42%).
- Civil is overstated (15% vs 10%).
- Mechanical is understated (15% vs 20%).

Viewers receive an incorrect impression of enrollment distribution.

## 3. Better chart

Use a **bar chart** with departments on the x-axis and student counts on the y-axis; bar lengths are easier to compare accurately than pie slices.

## 4. Effect on institutional planning

Incorrect proportions may lead to:

- Wrong faculty recruitment,
- Misallocation of classrooms and laboratories,
- Incorrect budgeting,
- Poor admission planning.

# Question 3: Unequal Bar Widths in Revenue Comparison

## 1. Graphical error

The chart uses **unequal bar widths** that increase from Product A to D.

## 2. How does it distort perception?

Viewers perceive **area (height × width)**, not height alone.

Approximate perceived areas:

- A:  $60 \times 1 = 60$
- D:  $105 \times 3 = 315$

D appears **more than 5 times larger** by area, while the actual ratio is only:

- $105 / 60 = 1.75$

### 3. Correct redesign principles

- Use **equal-width bars**,
- Keep **uniform spacing**,
- Start the Y-axis at **0**,
- Use a **consistent scale**,
- Avoid decorative width changes.

### 4. Consequences for investors

Investors may:

- Overvalue Product D,
- Underestimate Product A,
- Make poor investment decisions,
- Misjudge company performance and growth potential.

## Question 4: 3D Column Chart Distortion

### 1. Why can 3D charts mislead?

Perspective causes some bars to appear taller or larger depending on viewing angle. The visual size is affected by depth and shading, not just the data value.

### 2. Actual vs perceived difference

Actual values:

- Wind = 500 GWh
- Hydro = 550 GWh

Actual increase:

- $(550 - 500) / 500 \times 100 = \mathbf{10\%}$

Perceived increase from the chart:

- Approximately **40%**

The visual exaggeration is about **4 times** the true difference.

### 3. Better visualization

Use a **2D bar chart** or **horizontal bar chart** with a common baseline and no perspective effects.

#### 4. Importance in sustainability reporting

Sustainability reports are used by regulators, investors, and the public. Distorted charts can:

- Misrepresent environmental performance,
- Mislead policy decisions,
- Damage organizational credibility,
- Reduce trust in ESG reporting.

## Question 5: Bubble Chart with Incorrect Scaling

### 1. Why is radius scaling misleading?

If radius is proportional to cases, area becomes proportional to the **square** of the cases.

For A = 150 and E = 550:

- Radius ratio =  $550/150 \approx 3.67$
- Area ratio =  $3.67^2 \approx 13.4$

District E appears more than **13 times larger**, although it has only:

- $550/150 \approx 3.67$  **times** the cases.

### 2. Correct proportional sizing

Bubble **area** should represent the data value.

Use:

- $\text{Area} \propto \text{Cases}$
- $\text{Radius} \propto \sqrt{(\text{Cases})}$

This ensures visual area matches the numerical proportion.

### 3. Alternative visualization

Use a **bar chart** or **horizontal bar chart** for the five districts; exact comparisons are much easier.

### 4. Impact on public health resource allocation

An exaggerated bubble may cause officials to:

- Send excessive resources to District E,
- Neglect other districts,
- Misprioritize vaccination, staffing, or medical supplies,
- Distort outbreak severity assessment.

## Overall Conclusion

Across all five cases, the central principle is **truthful proportional representation**. Common sources of misleading visuals include truncated axes, incorrect pie proportions, unequal bar widths, 3D effects, and improper bubble scaling. Accurate visualization is essential because business, educational, financial, environmental, and public-health decisions often depend directly on graphical interpretation.